Digital Avionics — Homework 1

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Question 1

What are the advantages of digital avionics comparing the analog avionics?

Solution

Compared with analog avionics, digital avionics has the following advantages.

1. **Less weight and wiring.** One computer can perform several functions, and data buses allow different systems to share sensor information. This reduces the amount of separate hardware and cabling.
2. **Easier modification.** Control laws and display functions can be changed through software instead of changing the analog circuit design.
3. **More consistent processing.** Digital computation is less affected by analog component drift. Error checks can also detect corrupted data during transmission.
4. **Better integration.** Navigation, displays, and flight control can exchange information and work together. This also supports automation and reduces pilot workload.
5. **Fault detection and maintenance.** Built-in tests help locate faults. Digital monitoring and voting can detect a failed channel and allow the remaining channels to continue operating.

Question 2

Is there any drawback of the digital avionics? How to overcome the drawback in aircraft system engineering?

Solution

Yes. The main drawbacks and possible solutions are listed below.

1. **Aliasing:** If the sampling rate is too low, high-frequency signals can appear as false low-frequency signals. An anti-aliasing filter should be placed before the analog-to-digital converter, and the sampling rate should be sufficiently high.
2. **Quantization:** Digital values have limited resolution. A suitable number of bits and proper signal scaling reduce the resulting numerical error.
3. **Time delay and data staleness:** Processing and transmission take time, which can reduce control stability. Different computers may also compare data sampled at different times. These problems can be reduced by limiting delays, synchronizing the channels, and checking the age of the data.
4. **Software errors:** A software error can affect several computers using the same program. Careful verification and testing are required. An independent, dissimilar backup can help protect against common-mode failures.
5. **Shared equipment failures:** Failure of a shared computer, bus, or power supply may affect several functions. Redundant channels, independent power supplies, fault isolation, and backup operating modes can reduce this risk.

Therefore, aircraft system engineering must consider both the individual components and the complete system. Simply adding identical computers is not enough to protect against every type of failure.

Question 3

Introduce and compare the CNS/ATM and NAS (National Airspace System).

Solution

CNS/ATM stands for *Communications, Navigation and Surveillance / Air Traffic Management*. Its main functions are:

- **Communications:** transfer voice and data among aircraft and ground services.
- **Navigation:** determine position and support flight along an intended route.
- **Surveillance:** provide aircraft identity, position, and movement information.
- **ATM:** coordinate traffic services, airspace use, and traffic flow.

These functions use ground-based and satellite-based systems to support safe and efficient air traffic operations. ICAO helps coordinate their international implementation.

NAS stands for the U.S. *National Airspace System*. It includes airspace, airports, navigation facilities, air traffic services, operating rules, procedures, and personnel. The FAA manages civil air traffic within this system.

Both CNS/ATM and NAS aim to improve aviation safety and efficiency. The main difference is their scope: CNS/ATM describes the functions and technologies used for air traffic management, while NAS is the actual national system in the United States. CNS/ATM capabilities are used within the NAS, so the two concepts are related rather than separate alternatives.

Question 4

Every aircraft equips with voice communication by radio frequency. Many radio frequencies are used onboard for different avionics. Please show all the radio frequencies and the purposes of the belonging avionics.

Solution

The main avionics radio frequencies and their uses are summarized below. The equipment carried depends on the aircraft, and the assigned channels can vary between countries. The table gives operating bands or channel frequencies rather than every individual channel assignment.

System	Frequency or band	Purpose
HF communication	Assigned channels roughly within 2–23 MHz	Long-range air–ground voice/data; equipment may tune a wider range.
Civil VHF communication	118–137 MHz band	Line-of-sight ATC and operational voice/data. Not every channel is a voice channel.
Military UHF communication	225–400 MHz range, excluding reserved subbands	Military air–air and air–ground communications.
ACARS / VDL	Assigned VHF channels; VDL Mode 2 common signalling channel: 136.975 MHz	Operational/ATC data. ACARS can also use HF or SATCOM; it is not one universal RF frequency.
L-band aeronautical SATCOM	1545–1559 MHz downlink; 1646.5–1660.5 MHz uplink	Satellite voice/data; these are specified aeronautical mobile-satellite bands, not every provider's complete allocation.
Emergency voice	121.5 MHz; 243 MHz	Civil and military distress/guard communication.
ELT	406.0–406.1 MHz; commonly 121.5 MHz for homing	Distress alerting and search localization.
ADF / NDB	190–1750 kHz internationally; commonly 190–535 kHz in U.S. facilities	Aircraft receives beacon signals to determine bearing.
VOR	108.00–117.95 MHz, assigned channels	Bearing/radial guidance.
ILS localizer	Channel centres 108.10–111.95 MHz	Runway lateral guidance.
ILS glide slope	Channel centres 329.15–335.00 MHz	Vertical approach guidance; paired with localizer channels.
ILS marker beacon	75 MHz	Identifies fixed approach positions.

System	Frequency or band	Purpose
DME / TACAN	Within the 960–1215 MHz navigation band	DME measures slant range; TACAN also supplies bearing to suitably equipped military aircraft.
MLS	5030–5150 MHz allocation	Microwave precision-approach guidance.
GPS	L1: 1575.42 MHz; L2: 1227.60 MHz; L5: 1176.45 MHz	Satellite position, velocity, and time. These are GPS carriers; receiver use depends on its design and approval.
SSR / Mode A, C, S	1030 MHz interrogation; 1090 MHz reply	Cooperative identification, altitude reporting, and Mode S data.
TCAS / ACAS	1030 / 1090 MHz	Cooperative traffic detection and collision avoidance.
ADS-B	1090 MHz extended squitter; 978 MHz UAT	Broadcast aircraft state; UAT is a U.S. option below FL180, subject to operating requirements.
Radio altimeter	4200–4400 MHz	Height above terrain.
Airborne weather radar	Typically 9300–9500 MHz	Weather detection; 5350–5470 MHz is another airborne-radar allocation.
Doppler navigation radar	8750–8850 MHz or 13250–13400 MHz	Ground-relative velocity/drift measurement on equipped aircraft.

Communication systems exchange voice and data, navigation systems provide position or guidance, and surveillance systems help track aircraft. INS and wired data buses do not require their own external radio frequencies. Additional equipment, such as passenger connectivity or special-mission systems, may use other bands.

Question 5

Explain the relationships of Guidance, Navigation, and Control for an aircraft.

Solution

Navigation estimates the current position, velocity, attitude, and heading of the aircraft using sensors and navigation aids.

Guidance uses this information and the desired flight path to generate commands, such as the required heading, altitude, or bank angle.

Control operates the control surfaces and thrust to follow these commands. It also uses feedback to stabilize the aircraft against disturbances.

Their closed-loop relationship can be written as

$$\text{Desired path} \longrightarrow \text{Guidance} \longrightarrow \text{Control} \longrightarrow \text{Aircraft},$$

$$\text{Sensors} \longrightarrow \text{Navigation} \longrightarrow \text{Feedback to guidance and control.}$$

For example, if an aircraft moves away from its planned route, navigation detects the position error. Guidance determines the turn needed to return to the route, and control commands the surfaces to make the turn. The updated aircraft state is then fed back to guidance and control.

Common data for Questions 6–8

For Questions 6–8, the aircraft parameters from textbook §3.2.4 are used:

$$\begin{aligned} m &= 30{,}000 \text{ kg}, & S &= 75 \text{ m}^2, \\ C_{L,\max} &= 1.2, & \alpha_{\max} &= 15^\circ, \\ \rho_0 &= 1.225 \text{ kg/m}^3, & \rho_{30{,}000\,\text{ft}} &= 0.4583 \text{ kg/m}^3, \\ g &= 9.81 \text{ m/s}^2. \end{aligned}$$

Neglect the density variation from 0 to 1,000 ft. Following the example, assume a linear lift curve passing through the origin up to maximum lift:

$$C_L = \frac{C_{L,\max}}{\alpha_{\max}}\alpha = 0.08\alpha_{\text{deg}}.$$

Here α_{deg} is the angle in degrees. In this example, wing incidence refers to the aerodynamic angle of attack.

Question 6

What is the wing incidence, α, when flying straight and level at a speed of 80 m/s (160 knots approx.) at a height of 200 ft in the approach to the airfield?

Solution

For steady, unbanked, level flight, assuming thrust has no significant vertical component,

$$L = W = mg, \qquad L = qSC_L, \qquad q = \tfrac{1}{2}\rho V^2.$$

The weight and dynamic pressure are

$$W = 30{,}000(9.81) = 294{,}300 \text{ N}, \qquad q = \tfrac{1}{2}(1.225)(80)^2 = 3920 \text{ Pa}.$$

Therefore,

$$C_L = \frac{294{,}300}{3920(75)} \approx 1.001.$$

$$\boxed{\alpha = \frac{C_L}{0.08} \approx 12.5^\circ.}$$

The required angle is below 15°, so the lift requirement is achievable within the assumed lift-curve range.

Question 7

What is the corresponding wing incidence when flying straight and level at a speed of 200 m/s (400 knots approx.) at a height of 1,000 ft?

Note: Neglect change in density 0–1,000 ft *for simplicity.*

Solution

The density, weight, and wing area remain the same. Thus

$$q = \tfrac{1}{2}(1.225)(200)^2 = 24{,}500 \text{ Pa},$$

$$C_L = \frac{mg}{qS} = \frac{294{,}300}{24{,}500(75)} \approx 0.1602,$$

and

$$\boxed{\alpha = \frac{C_L}{0.08} \approx 2.0^\circ.}$$

As a cross-check, L is fixed and C_L is proportional to α, so

$$\alpha_7 = \alpha_6 \left(\frac{V_6}{V_7}\right)^2 \approx 12.5^\circ \left(\frac{80}{200}\right)^2 = 2.0^\circ.$$

The higher speed produces greater dynamic pressure, so less angle of attack is needed to support the same weight.

Question 8

What is the maximum normal acceleration that can be achieved at a height of 30,000 ft when flying at a speed of 225 m/s (450 knots)? Neglect compressibility effects and assume adequate thrust is available to counteract the increase in drag and maintain speed of 450 knots at $\alpha_{\max}$.

Solution

Using $V = 225\,\text{m/s}$ and $C_{L,\max} = 1.2$ at $\alpha_{\max} = 15^\circ$,

$$q = \tfrac{1}{2}(0.4583)(225)^2 \approx 11{,}600.7 \text{ Pa},$$

$$L_{\max} = qSC_{L,\max} \approx 11{,}600.7(75)(1.2) \approx 1.0441 \times 10^6 \text{ N}.$$

For a wings-level pull-up starting from a horizontal flight path, the net normal force is lift minus weight. Therefore,

$$ma_{n,\max} = L_{\max} - mg,$$

$$\begin{aligned} a_{n,\max} &= \frac{L_{\max} - mg}{m} \\ &\approx \frac{1.0441 \times 10^6 - 294{,}300}{30{,}000} \\ &\approx 25.0 \text{ m/s}^2. \end{aligned}$$

Hence,

$$\boxed{a_{n,\max} \approx 25.0 \text{ m/s}^2 \approx 2.55g.}$$

This is approximately $2.5g$ using the rounding in the textbook. The corresponding load factor is

$$n_{\max} = \frac{L_{\max}}{W} \approx 3.55.$$

The load factor is different from the net normal acceleration because gravity must be subtracted: $a_n = (n-1)g$. This calculation neglects compressibility and assumes sufficient thrust, as specified in the question.